

Putnam Seminar #1

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Let's get something straight: the Putnam is **hard**. Median score is usually 1, maybe 2. A single training session will maybe improve your score by 0.00000000001%. So, if you're here, it's because you

- Like solving problems
- Like math
- Want to challenge yourself
- Want to meet people with similar interests

Still here? Then let's dive in.

Problems

Today we'll cover a problem in combinatorics.

Putnam 2024, B1

Let n and k be positive integers. The square in the i -th row and j -th column of an n -by- n grid contains the number $i + j - k$. For which n and k is it possible to select n squares from the grid, no two in the same row or column, such that the numbers contained in the selected squares are exactly $1, 2, \dots, n$?

Before we dive into the problem, let's uncover the ideas behind it.

Appetizer #1

An $n \times n$ Latin square A is *symmetric* if it is symmetric around the main diagonal, that is, $A_{ij} = A_{ji}$ for all i and j . It is *idempotent* if every symbol appears on the main diagonal. Show that if A is both symmetric and idempotent, then n is odd. Find a 5×5 symmetric, idempotent Latin square.

Solution. We will count the number of 1's in the Latin square. On one hand, we know there are n 1's on the square. On the other hand, let k be the number of 1's below the main diagonal. Note that there are k ones above the main diagonal (since A is symmetric) and exactly 1 one on the diagonal (since A is idempotent). Then $n = 2k + 1$, i.e., n is odd.

A construction for $n = 5$ is shown below (the addition table modulo 5) ■

Appetizer #2

A *transversal* on an $n \times n$ Latin square is a set of n cells, no two in the same row or column, such that there is a cell containing each symbol. Determine for which values of n we can find a transversal on the Latin square corresponding to the addition table of $\mathbb{Z}_n$.

Solution. We will show that we can find transversals if and only if n is odd.

First, suppose that there exists a transversal on the addition table of $\mathbb{Z}_n$. Let (i, j) denote the cell on the i -th row and j -th column, where $0 \leq i \leq j \leq n-1$. Then the number in the cell (i, j) is precisely $i + j \pmod{n}$. Let $(a_1, b_1), \dots, (a_n, b_n)$ denote the cells of the transversal. Then the sets $\{a_1, \dots, a_n\}$ and $\{b_1, \dots, b_n\}$ are both equal to the set $\{0, 1, \dots, n-1\}$. Furthermore, the numbers on these cells are

$$a_1 + b_1 \pmod{n}, \dots, a_n + b_n \pmod{n}$$

which are precisely the numbers $0, 1, \dots, n-1$ in $\mathbb{Z}_n$.

Then

$$\begin{aligned} 0 + 1 + \dots + n-1 &\equiv (a_1 + b_1) + \dots + (a_n + b_n) \pmod{n} \\ \frac{n(n-1)}{2} &\equiv (a_1 + \dots + a_n) + (b_1 + \dots + b_n) \pmod{n} \\ \frac{n(n-1)}{2} &\equiv \frac{n(n-1)}{2} + \frac{n(n-1)}{2} \pmod{n} \\ \frac{n(n-1)}{2} &\equiv 0 \pmod{n}. \end{aligned}$$

That is, n divides $\frac{n(n-1)}{2}$, which implies that $\frac{n-1}{2}$ is an integer. Therefore, n is odd.

To find a construction for odd n , it suffices to pick the cells on the main diagonal; i.e., the cells

$$(0, 0), (1, 1), \dots, (n-1, n-1).$$

The exact proof that these cells contain the numbers $0, 1, \dots, n-1$ is left to the reader. ■

Main course: Putnam 2024, B1

Let n and k be positive integers. The square in the i -th row and j -th column of an n -by- n grid contains the number $i + j - k$. For which n and k is it possible to select n squares from the grid, no two in the same row or column, such that the numbers contained in the selected squares are exactly $1, 2, \dots, n$?

Solution. We will show that the desired construction only exists when n is odd and $k = \frac{n+1}{2}$.

First, suppose that n and k are such that the desired construction exists. Let $(a_1, b_1), \dots, (a_n, b_n)$ be cells of the grid, where no two are in the same row or column. Then the sets $\{a_1, \dots, a_n\}$ and $\{b_1, \dots, b_n\}$ both equal the set $\{1, \dots, n\}$. Note that the numbers in each cell are

$$a_1 + b_1 - k, \dots, a_n + b_n - k.$$

If these n numbers happen to equal the numbers $1, 2, \dots, n$ in some order, then

$$\begin{aligned} \frac{n(n+1)}{2} &= \sum_{i=1}^n (a_i + b_i - k) \\ &= \sum_{i=1}^n a_i + \sum_{i=1}^n b_i - \sum_{i=1}^n k \\ &= \frac{n(n+1)}{2} + \frac{n(n+1)}{2} - nk. \end{aligned}$$

Then $k = \frac{n+1}{2}$. Since k is an integer, n must be even.

When n is odd and $k = \frac{n+1}{2}$, the following construction works:

$$\left(1, \frac{n+1}{2}\right), \left(2, \frac{n+3}{2}\right), \dots, \left(\frac{n+1}{2}, n\right)$$

$$\left(\frac{n+3}{2}, 1\right), \left(\frac{n+5}{2}, 2\right), \dots, \left(n, \frac{n-1}{2}\right)$$

The exact proof is left to the reader. ■

Preparation tips

- Past Putnam problems, especially A1/B1, A2/B2. Focus on elementary algebra, number theory, combinatorics, calculus.
- Practice proof-writing skills. Go to easier problems whenever you need: AMC 12 or AIME problems. Even textbook exercises work.
- Learn new topics as you go. You can't possibly cover *everything* you need, because *everything is fair game*.
- If you're still hungry for problems, you can look into high school math contests. Even beyond the IMO and USAMO, there are contests from all around the world, each with their own flavor.

Further resources

- Kiran Kedlaya's Putnam Archive: <https://kskedlaya.org/putnam-archive/>
- Past Putnam seminars by Yufei Zhao (ocw.mit.edu/courses/18-a34-mathematical-problem-solving-putnam-seminar-fall-2018/pages/) and Po-Shen Loh: <https://www.math.cmu.edu/~ploh/2024-putnam.shtml>
- Olympiad preparation handouts by Evan Chen: <https://web.evanchen.cc/olympiad.html> → oriented towards USAMO and IMO
- Art of Problem Solving forums: https://artofproblemsolving.com/community/c3249_putnam (Putnam forum)
- Putnam and Beyond by Titu Andreescu and Razvan Gelca: <https://link.springer.com/book/10.1007/978-3-319-58988-6>